

# Measuring Dynamic Trust for Human-AI Complementarity: Toward a Longitudinal Evaluation Framework

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## Abstract

Gonzalez et al. (2026) argue that evaluating human-AI teaming requires methods that assess trust calibration and adaptation across interactions, not just single-turn accuracy, yet such methods remain underdeveloped. Complementarity cannot be assumed. Recent evidence shows that human-AI teams frequently perform worse than the stronger individual contributor (Vaccaro, Almaatouq, & Malone, 2024), motivating longitudinal measures that explain when calibrated trust actually produces complementarity. We argue that longitudinal trust calibration provides a behavioral lens for evaluating dynamic alignment between humans and AI.

We present a longitudinal evaluation framework integrating behavioral and subjective trust measures from human-human team science and human-machine trust research into a common approach for evaluating dynamic human-AI teams. Human-human team science contributes established behavioral markers of teamwork, including grounding and repair in communication, transactive memory, and backup behavior, that provide candidate behavioral indicators of trust-related processes during collaboration (Clark & Brennan, 1991; Lewis, 2003; Porter et al., 2003). Human-machine trust research complements these measures with prediction of AI error boundaries, reliance and override behavior, and longitudinal trust ratings that capture calibration across repeated interactions (Bansal et al., 2019; McGuirl & Sarter, 2006; Yang, Schemanske, & Searle, 2021).

Building on these complementary traditions, we define dynamic trust as trust continuously updated through interaction and expressed through observable reliance. The framework organizes trust into a four-stage feedback loop consisting of observable behavior, trust calibration, adaptive reliance and task allocation, and updated behavior across successive interactions. It emphasizes behavioral measures from Crew Resource Management marker systems (Salas et al., 2006), while treating convergence and divergence with subjective trust ratings as diagnostically meaningful.

The framework provides a common language for evaluating how calibrated trust contributes to dynamic alignment and human-AI complementarity over time. We seek workshop feedback on which trust mechanisms, and which combinations of behavioral and subjective measures, best transfer to and capture complementarity in human-AI teams.

# 1. Introduction and Gap

Human-AI teaming is often evaluated through single-turn accuracy or task outcome measures, which capture how well a team performed on a given trial but say little about how the team arrived at that performance or whether it will hold up as conditions change. Gonzalez et al. (2026) identify this as a central gap in the science of human-AI teaming for decision making: evaluating complementarity requires methods that assess trust calibration and adaptation across a sequence of interactions, not a single decision point.

This gap matters because complementarity is not a given property of pairing a human with an AI system. Vaccaro, Almaatouq, and Malone (2024) report that in a systematic review and meta-analysis, human-AI teams frequently underperform relative to the stronger of the two individual contributors, meaning the team does worse than either the human or the AI would have done alone. If complementarity cannot be assumed, evaluators need methods that explain when and why calibrated trust translates into performance gains, and when it does not. A single accuracy measure at one point in a task sequence cannot answer that question, because it collapses a dynamic process into a static endpoint.

Trust in AI-enabled systems has become a substantial research area in its own right, and recent systematic reviews document a wide range of proposed constructs and measurement approaches (Bach et al., 2024; Kohn et al., 2021). This body of work is useful but fragmented: a given study may draw on subjective trust scales, behavioral reliance measures, or physiological indicators, with little common structure connecting them across studies or domains. The present framework does not attempt to catalog this literature exhaustively. Instead, it selects a small set of behavioral markers with an existing validated structure in team science, and organizes them longitudinally.

This paper describes a longitudinal evaluation framework intended to fill that gap. We treat trust as an evolving state, continuously updated through interaction and expressed through observable reliance, rather than a fixed attitude reported once. The framework structures this evolution as a four-stage feedback loop, discussed in Section 3, drawing on measurement traditions from two adjacent literatures: human-human team science and human-machine trust research, discussed in Section 2.

Doing so requires separating three constructs that are often conflated in human-AI evaluation: capability estimation, trust, and reliance. Capability estimation concerns what the human believes the AI can do in a given situation. Trust concerns the human's willingness to accept vulnerability based on that belief, that is, to act on the AI's output without independently verifying it. Reliance is the resulting observable behavior, the decision to accept, reject, override, or modify the AI's output. Calibration, in turn, concerns the correspondence between capability estimation and the AI's actual demonstrated capability. In this framework, trust calibration is the mechanism connecting capability estimation to adaptive reliance: an operator who accurately estimates AI capability and whose trust tracks that estimate should show reliance that varies appropriately with AI performance, while an operator with poorly calibrated trust should not. This distinction connects directly to the complementarity question raised by Gonzalez et al. (2026): a

team can only realize complementary capabilities if the human's reliance decisions track what the AI can actually do, rather than a fixed overall impression of the AI's trustworthiness.

## 2. Measurement Framework: NOTECHS Extended to Trust

The NOTECHS framework was developed to evaluate behavioral markers of non-technical skills in Crew Resource Management training, organized around categories such as cooperation, management, situation awareness, and decision-making (Salas et al., 2006). Matessa (2017) extended this marker structure to human-AI collaboration, applying the same category structure to interactions between human operators and AI or automated systems. The present framework extends this line of work a second time, using the NOTECHS categories to organize measures relevant to trust between humans and between a human and an AI system.

To be clear about what this extension claims: the present framework does not treat the NOTECHS categories as dimensions of trust. Cooperation, management, situation awareness, and decision-making are domains of teamwork behavior, not components of trust itself. Rather, the framework uses them as an organizing taxonomy for locating observable behaviors through which trust may be expressed, calibrated, and revised during teamwork.

Organizing trust measures this way serves two purposes. First, it draws on behavioral marker systems already established in team science, rather than proposing trust measures from scratch. Second, it makes the parallels and departures between human-human trust and human-machine trust explicit, which is useful for evaluators deciding which measures transfer to human-AI settings and which need to be adapted or replaced.

This extension also builds on prior work applying team-cognition constructs to synthetic and AI teammates specifically, including the synthetic teammate project (Ball et al., 2010) and models of interactive team cognition that treat cognitive activity as distributed across human and artificial team members rather than located in any one individual (Cooke, Gorman, Myers, & Duran, 2013).

### 2.1 Human-Human Trust: Three Ways Trust Is Operationalized

Before cataloging specific markers, it is useful to distinguish three ways trust has been operationalized in the human-human literature, since each carries different implications for what a corresponding human-machine measure should look like:

- **Direct behavioral operationalizations of trusting behavior**, in which the act of trusting is itself the measured behavior: investing money in an economic trust game (Johnson & Mislin, 2011), delegating responsibility to another party, or choosing whether to rely on someone's advice.
- **Observable team behaviors theorized to express trust**, such as information sharing, monitoring, and backup behavior, which are not trust itself but are argued to reflect an underlying trust state.

- **Interaction and process measures that provide behavioral evidence of trust**, such as communication, repair, and coordination, which capture how trust is enacted moment to moment rather than as a single decision.

Within these categories, the NOTECHS structure organizes trust-relevant behavior as follows:

- **Cooperation.** Monitoring teammates' actions and states (McIntyre & Salas, 1995), the tendency to withhold information from teammates, and backing up behavior, in which team members compensate for one another's workload or errors (Porter et al., 2003).
- **Management.** Delegation of tasks or authority to teammates (Gur & Bjørnskov, 2017), and task allocation decisions about who performs which part of a joint task, including entrustment decisions in supervisory training contexts (Ten Cate, 2005).
- **Situation Awareness.** Verification behavior, in which a team member checks another's claims rather than simply relying on them, which can itself be a form of prosocial deception management (Levine & Schweitzer, 2015); grounding and repair, the process by which communicating parties establish and correct shared understanding (Clark & Brennan, 1991); and transactive memory, the shared system by which team members track who knows what and rely on one another's distinct expertise (Wegner, 1987; Lewis, 2003).
- **Decision-making.** Reliance on a teammate's input, studied directly through economic trust games (Johnson & Mislin, 2011), and advice acceptance in judge-advisor paradigms, where one party revises a judgment after receiving another's recommendation (Snizek & Van Swol, 2001).

A separate strand of this literature addresses how trust is established before a team has an interaction history to draw on. Swift trust describes the trust that temporary teams extend to one another at formation, based on role and category expectations rather than accumulated experience (Meyerson, Weick, & Kramer, 1996). This is directly relevant to Stage 1 of the framework below, where an operator's very first observation of an AI system necessarily occurs before any track record has been established.

## 2.2 Grounding as a Candidate Behavioral Indicator of Trust Calibration

We propose that grounding behavior may provide an observable behavioral signature of trust calibration, extending Clark and Brennan's (1991) grounding framework beyond its original scope as a property of communication. This proposition is not established by the grounding literature itself; it follows from combining grounding theory with the dynamic trust literature reviewed above, and should be treated as a hypothesis to be tested rather than a validated measure. If the proposition holds, the amount and type of grounding behavior in human-AI teams could reveal whether the human is actively calibrating trust in the AI, rather than simply holding a fixed level of trust throughout an interaction. Table 1 lists candidate grounding behaviors and the calibration states they may indicate.

**Table 1. Grounding behaviors and possible trust interpretations**

| Behavior                                | Possible interpretation          |
|-----------------------------------------|----------------------------------|
| Immediate acceptance                    | High trust or overtrust          |
| Frequent clarification                  | Active calibration               |
| Confirmations                           | Verification before reliance     |
| Repair after disagreement               | Maintaining calibrated trust     |
| Ignoring recommendations                | Distrust or justified skepticism |
| No questions despite repeated AI errors | Possible overtrust               |

For example, increasing clarification after AI errors, followed by a return to efficient acknowledgments as the AI proves reliable, may indicate healthy trust adaptation: grounding effort rises when calibration is genuinely uncertain and falls once the operator has established a reliable read on the AI's capability. Conversely, immediate acceptance despite repeated errors, or persistent rejection despite reliable performance, could signal miscalibration, since in both cases the operator's grounding behavior fails to track the AI's actual demonstrated performance. Treating grounding this way turns it from a purely communicative construct into a candidate observable measure for Stages 1 and 4 of the framework in Section 3.

## 2.3 Trust-Relevant Team Behaviors

Beyond grounding specifically, the human-human team science literature has identified a broader set of behaviors associated with high or well-functioning trust. Table 2 summarizes these markers.

**Table 2. Trust-Relevant behavioral markers**

| Marker               | Definition                                                                     | Source                 |
|----------------------|--------------------------------------------------------------------------------|------------------------|
| Grounding and repair | Establishing and restoring mutual understanding through communication          | Clark & Brennan (1991) |
| Transactive memory   | Coordinated reliance on distributed, specialized knowledge across team members | Lewis (2003)           |
| Backup behavior      | Stepping in to help a teammate complete a task or correct performance          | Porter et al. (2003)   |

|                                                   |                                                                                                                 |                                                          |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Mutual performance monitoring                     | Tracking teammates' actions and state during task execution, typically preceding backup behavior                | McIntyre & Salas (1995)                                  |
| Information sharing and acceptance of influence   | Openly sharing information and accepting input from teammates rather than withholding it                        | Zand (1972)                                              |
| Voice and disclosure                              | Speaking up about errors, asking for help, and raising concerns without concealment                             | Edmondson (1999)                                         |
| Constructive task conflict                        | Engaging in open task debate without it escalating into relationship conflict or defensiveness                  | Simons & Peterson (2000)                                 |
| Coordinated action without explicit communication | Convergence in team members' task-relevant expectations that shows up as smooth, low-communication coordination | Mathieu, Heffner, Goodwin, Salas, & Cannon-Bowers (2000) |

These markers cluster into two families: monitoring, backup, and coordination behaviors on one side, and disclosure, information-sharing, and voice behaviors on the other (Costa, Fulmer, & Anderson, 2018). This clustering maps loosely onto the distinction between cognitive-based and affective-based trust (McAllister, 1995), with the monitoring and coordination family tracking more closely with a rational assessment of a teammate's competence and reliability, and the disclosure and voice family tracking more closely with a relational willingness to be vulnerable. For evaluators building a coding scheme rather than a narrative review, this two-family structure may be a more tractable way to organize markers than a flat list.

## 2.4 Distrust-Related Behaviors

Low-trust behaviors are less directly studied as their own category in this literature. Most papers frame trust on a continuum and describe its absence rather than validating separate low-trust measures. Several papers do, however, isolate distinct behavioral signatures of low or eroding trust, summarized in Table 3.

**Table 3. Distrust-Related behavioral markers**

| Marker                                            | Definition                                                                            | Source      |
|---------------------------------------------------|---------------------------------------------------------------------------------------|-------------|
| Withholding information and guarded communication | Restricting information flow and hedging disclosure to protect against perceived risk | Zand (1972) |

|                                             |                                                                                                                  |                                                                     |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Relationship conflict and defensiveness     | Task disagreement escalating into personal friction, accompanied by defensive posturing                          | Simons & Peterson (2000)                                            |
| Excessive or redundant monitoring           | Checking behavior that exceeds what the task requires, read as a substitute for trust rather than support for it | Langfred (2004);<br>Bijlsma-Frankema, De Jong, & Van de Bunt (2008) |
| Organizational silence and self-censorship  | Deliberately withholding ideas, concerns, or dissent from those in a position to act on them                     | Morrison & Milliken (2000)                                          |
| Territorial and knowledge-hoarding behavior | Marking out and defending exclusive control over information, tasks, or resources rather than sharing them       | Brown, Crossley, & Robinson (2014)                                  |
| Reduced help-seeking and error concealment  | The inverse of the voice marker in Table 2: not asking for help and not disclosing mistakes                      | Edmondson (1999)                                                    |

A methodological caveat applies to the monitoring marker in particular. Several sources (Langfred, 2004; Bijlsma-Frankema et al., 2008) frame the trust-monitoring relationship as curvilinear or context-dependent, rather than establishing that low monitoring simply indicates high trust and high monitoring simply indicates low trust. Costa et al. (2018) make this point directly: whether monitoring reads as distrust depends on whether it is expected in the given context and how it is perceived by the party being monitored. For a coding scheme, this ambiguity should be named explicitly rather than treating monitoring frequency alone as a stable low-trust indicator.

## 2.5 Trust and Task Allocation Across Team Types

Trust's role in task allocation has been studied across several kinds of teams, with the relationship growing more formalized as the non-human partner becomes more autonomous.

In human-human teaming, interpersonal trust supports delegation, specialization, and division of labor (Meier, Stephenson, & Perkowski, 2019). Langfred (2004, 2007) offers an important complication rather than simply supporting evidence: high trust combined with high individual autonomy can reduce monitoring to the point that it hurts performance, showing that more trust and less oversight is not uniformly beneficial for task allocation outcomes.

In human-automation settings, operator trust influences the allocation of tasks to automation (Lewandowsky, Mundy, & Tan, 2000). Lewandowsky et al. directly compare human and automated partners within the same experimental paradigm and find that the trust-allocation relationship is qualitatively identical but quantitatively attenuated for automated partners relative

to human partners, meaning the same basic pattern holds but is weaker when the partner is a machine.

In human-robot teaming, trust has moved beyond a variable that is merely measured to one that is explicitly incorporated into task-allocation algorithms. Ali, Azevedo-Sa, Tilbury, and Robert (2021, 2022) formalize trust-based allocation for heterogeneous human-robot teams, and related work extends this to mutual-trust-based subtask allocation in manufacturing (Rahman & Wang, 2018) and to multi-human multi-robot task allocation settings (Obi, Wang, Jo, & Min, 2025). This body of work is useful for the present framework because it demonstrates that the trust-allocation relationship can be operationalized computationally, not just measured descriptively.

In human-AI teaming specifically, emerging work extends this to dynamic allocation, adaptive autonomy, and AI-mediated work design, but empirical evidence remains comparatively limited relative to the human-robot literature. Recent work identifies trust and human agency explicitly as open issues in dynamic human-AI task allocation. Wilder, Horvitz, and Kamar (2020) learn when a task should be handled by a human versus an AI system based on the relative capabilities of each, demonstrating improved team performance from this capability-based allocation approach without directly modeling trust as the allocation mechanism. Fügner, Grahl, Gupta, and Ketter (2022) examine the cognitive challenges involved in productive delegation between humans and AI, closing the loop between the task-allocation literature reviewed here and the present framework's Stage 3.

## 2.6 Human-Machine Trust

The human-machine trust literature offers measures within the same four NOTECHS categories, developed specifically for interactions with automated or AI systems:

- **Cooperation.** Use of AI-provided confidence information when making a decision (McGuirl & Sarter, 2006), and dynamic trust adjustment, in which an operator's trust shifts in response to the AI's demonstrated performance over successive interactions (Yang, Schemanske, & Searle, 2021).
- **Management.** Delegation to an AI system, examined as a function of the contextual information available to the human at the time of the decision (Spitzer et al., 2025), and task allocation between human and automation (Lewandowsky, Mundy, & Tan, 2000).
- **Situation Awareness.** Predicting the boundaries of an AI system's reliable performance, that is, anticipating where it is likely to succeed or fail (Bansal et al., 2019), and verification of AI output, examined in multi-agent supervisory contexts (Walliser, de Visser, & Shaw, 2016).
- **Decision-making.** Over-reliance and under-reliance on automation, the classically studied failure modes in which operators either trust automation more or less than its actual reliability warrants (Parasuraman & Riley, 1997), and recommendation modification, where a human adjusts rather than simply accepts or rejects an AI's suggestion (Liu et al., 2023).

The parallel structure across Sections 2.1 and 2.6 is intentional. It allows an evaluator to ask, for a given human-AI team, which human-human trust measures have a direct human-machine counterpart already validated in the literature, and where a category is thinner on the human-machine side and might benefit from adapting the human-human measure directly.

### 3. Dynamic Trust Framework: A Four-Stage Feedback Loop

The measures described above are organized into a four-stage cycle describing how trust is built, applied, and revised across a sequence of interactions between a human and an AI system. Each stage maps to specific NOTECHS-derived measures, and the stages feed into one another in a closed loop rather than a linear sequence. Framed in terms of the CRM-derived behavioral markers reviewed in Section 2, the loop can be summarized as: observable CRM behavior (communication, monitoring, coordination, verification, delegation) feeding into a trust calibration judgment, which drives an adaptive reliance or task-allocation decision (accept, reject, override, delegate, reallocate), which produces updated CRM behavior following the observed AI success, failure, or repair.

**Stage 1: Observe.** Before any decision to rely on the AI is made in the current cycle, the operator forms a read of the situation. This stage is not trust-free: operators enter each cycle with trust carried forward from previous interactions, along with task and system expectations, consistent with the swift trust that can precede any interaction history at all (Meyerson, Weick, & Kramer, 1996). The purpose of this stage is to acquire new evidence against which those existing expectations can be evaluated, prior to any allocation decision in the current cycle. It is measured through situation awareness at the levels of perception and comprehension, that is, noticing the relevant elements of the current state and understanding what they mean. Observable CRM markers at this stage include communication, monitoring, and initial verification behavior.

**Stage 2: Calibrate.** The operator projects forward from the current situation to anticipate how the AI is likely to perform under these specific conditions. This is measured through the projection component of situation awareness, closely related to the human-machine trust measure of predicting AI error boundaries (Bansal et al., 2019): given what the operator currently understands about the situation, is this a context in which the AI can be expected to perform reliably? This stage combines subjective trust judgment with an inferred trust state relative to the AI's observed capability.

**Stage 3: Allocate and rely.** Based on the calibration formed in Stage 2, the operator makes a management and decision-making choice about whether and how much to rely on the AI: accept, reject, override, delegate, or reallocate the task. The framework specifies a directional prediction here: good calibration paired with good AI performance should produce increased use of the AI, while good calibration paired with poor AI performance should produce reduced use, that is, the operator correctly identifies the AI as unreliable in this instance and relies on it less. This directional prediction is consistent with the broader concept of appropriate reliance, in

which trust is treated as well-calibrated not when it is uniformly high, but when it tracks the automation's actual capability across situations (Lee & See, 2004). This stage draws on delegation and task allocation measures (Spitzer et al., 2025; Lewandowsky et al., 2000) together with reliance and override measures (Parasuraman & Riley, 1997), and connects to the trust-and-allocation literature reviewed in Section 2.5.

**Stage 4: Observe consequences.** After acting on the Stage 3 decision, the operator observes the outcome, whether the AI performed as expected or not, and this outcome feeds back into the trust the operator carries forward. This stage is measured through cooperation markers, particularly dynamic trust adjustment (Yang, Schemanske, & Searle, 2021) and use of AI confidence information (McGuirl & Sarter, 2006), capturing how the observed outcome updates the operator's trust going into the next cycle. This mirrors findings that both an AI system's demonstrated accuracy (Yin, Wortman Vaughan, & Wallach, 2019) and the confidence or explanation information it presents alongside its output (Zhang, Liao, & Bellamy, 2020) shape how much trust an operator carries forward into subsequent decisions. Changes in grounding behavior (Table 1) and in the CRM markers reviewed in Section 2 following AI successes, failures, and repair events are the observable evidence at this stage.

The loop then returns to Stage 1, with the trust level carried forward from Stage 4 becoming part of the prior trust the operator brings into the next cycle's observation. The full loop is therefore: prior trust and expectations, observe, calibrate, allocate and rely, observe consequences, updated trust. This closure is the central theoretical claim of the framework: trust calibration is not a single judgment formed once, but a state that is continuously revised across repeated cycles of observation, calibration, allocation, and consequence, consistent with broader models of team adaptation as an ongoing cycle of monitoring and adjustment rather than a one-time achievement (Burke, Stagl, Salas, Pierce, & Kendall, 2006).

### 3.1 Illustrative Example

To make the loop concrete, consider a hypothetical example: an operator supervising an AI-based conflict-detection tool during uncrewed aircraft system (UAS) flights beyond visual line of sight. The AI flags potential traffic conflicts from sensor data across a series of flights. The example is illustrative rather than an empirical validation of the framework.

In one cycle, the AI performs well: the operator verifies an AI-generated alert against raw sensor data and finds it accurate (Stage 1); based on current conditions, the operator judges the situation to fall within the AI's known reliable range (Stage 2); the operator delegates ongoing monitoring of routine traffic to the AI (Stage 3, consistent with the good-calibration/good-performance branch of the framework); the AI's next alert is also correct, and the operator's trust rises, using the AI's confidence score to gauge response urgency, while grounding behavior settles into efficient acknowledgment rather than repeated clarification (Stage 4). That elevated trust becomes the starting point for the next cycle.

In a later cycle, conditions degrade, for example due to sensor dropout in poor weather. The operator's Stage 1 verification catches a conflict the AI missed. Stage 2 calibration revises the

AI's reliability downward for these conditions. Stage 3 reliance decreases, consistent with the good-calibration/poor-performance branch: the operator pulls monitoring back to manual cross-checking rather than continuing to delegate. Stage 4 registers a drop in trust and an increase in clarification and verification behavior, which the operator carries into subsequent flights until the AI demonstrates reliable performance again.

This pairing illustrates the framework's core claim: the same operator, the same named measures at each stage, and the same four-stage structure produce opposite reliance outcomes depending on how well the AI performs, which is what distinguishes trust calibration from a fixed trust level.

## **4. Literature Gap: Toward a Full-Cycle Test of CRM Behavioral Markers**

Recent work has made progress on individual pieces of the loop described in Section 3, but these components have generally been studied separately. Trust-dynamics research has modeled continuous trust updating from trial-by-trial reliance decisions (Yang, Schemanske, & Searle, 2021; Guo & Yang, 2021), and task-allocation research has modeled trust-based allocation computationally (Ali et al., 2022) or measured trust's effect on one-shot allocation decisions (Lewandowsky, Mundy, & Tan, 2000). The literature reviewed here suggests an opportunity to integrate these separately studied components, trust calibration, behavioral task allocation, resulting performance, and subsequent trust updating, within a common longitudinal evaluation framework using continuous behavioral measures rather than periodic self-report alone. This is consistent with recent calls for more rigorous, practically grounded approaches to evaluating calibrated trust (McDermott & Brink, 2019) and with recent surveys finding that trust calibration measurement for automated systems remains fragmented across methods and lacks a common evaluation standard (Wischnewski, Krämer, & Müller, 2023).

We frame this gap in terms of the CRM-derived behavioral markers reviewed in Section 2, rather than treating information sharing and monitoring as separate, unrelated gaps. Human-machine trust research has primarily operationalized trust through subjective ratings and reliance behaviors. By contrast, human-human teamwork research has developed richer behavioral markers of trust-relevant interaction, including information sharing, monitoring, cross-checking, coordination, assertion, clarification, and mutual support. Although some of these behaviors, especially monitoring, communication, coordination, and information sharing, do appear individually in human-automation and human-AI teaming research, they have not been systematically evaluated together as a multidimensional behavioral signature of dynamic human-machine trust. The claim here is that these markers have not been systematically conceptualized and validated for this purpose, not that they are entirely absent from the literature.

This framing is theoretically useful because CRM markers sit between trust and reliance in the causal chain: trust shapes interactional behavior, which in turn shapes reliance and task allocation. A highly trusted human teammate might proactively share information, provide

relevant warnings, communicate uncertainty, coordinate rather than act independently, delegate responsibility, and reduce unnecessary checking. After a trust violation, the same teammate might show increased cross-checking, more verification, more questioning, reduced delegation, increased monitoring, and more explicit confirmation. These are behavioral signatures of trust updating, rather than outcome measures such as whether an AI recommendation was simply accepted.

This framing also addresses a measurement problem with reliance as a standalone indicator. Acceptance of an AI recommendation is ambiguous on its own: it could reflect trust, habit, workload, time pressure, lack of alternatives, or organizational policy, among other explanations. A pattern such as an AI error followed by increased questioning, cross-checking, monitoring, and reduced delegation provides substantially richer evidence that the operator's relationship with the system has actually changed, compared to observing the acceptance or rejection decision alone.

Mapped onto the four-stage framework in Section 3, this yields a specific empirical question worth testing directly: do CRM behavioral markers provide earlier or more sensitive indicators of trust updating than subjective trust ratings or reliance measures alone? This question turns the CRM literature from a source of additional candidate measures into a testable proposition about how dynamic human-machine trust should be measured.

## **5. Discussion and Open Questions**

The framework is proposed as a common language for evaluators, not a finished measurement instrument. Several open questions follow directly from the framework's structure and the literature gap identified above, and are the focus of the workshop discussion this paper accompanies.

First, which combinations of behavioral and subjective measures best transfer to, and capture, complementarity in human-AI teams? The framework draws primarily on behavioral markers from Crew Resource Management systems (Salas et al., 2006), but human-machine trust research also has a well-established tradition of subjective trust scales (Jian, Bisantz, & Drury, 2000). The framework treats convergence and divergence between behavioral measures and subjective ratings as diagnostically meaningful in itself: an operator who reports high trust but shows reduced reliance, or vice versa, may reveal something about trust calibration that neither measure alone would show. The empirical question raised in Section 4, whether CRM markers detect trust updating earlier or more sensitively than subjective ratings or reliance alone, is a specific version of this broader question.

Second, which trust mechanisms generalize across domains? The example in Section 3.1 is drawn from an aerospace supervisory context, consistent with the framework's origins in Crew Resource Management and prior work applying NOTECRS-based markers to human-AI collaboration in aerospace settings (Matessa, 2017). Whether the same four-stage structure, and the same directional prediction in Stage 3, holds in other domains, for example clinical

decision support, financial advising, or content moderation, is untested. Trust repair processes following a failure (de Visser, Pak, & Shaw, 2018; Kim, Dirks, & Cooper, 2009), longitudinal trust calibration in other human-machine team settings (de Visser et al., 2020), and computational approaches to modeling and predicting moment-to-moment trust dynamics (Yang, Guo, & Schemanske, 2022) suggest the underlying mechanisms may generalize, but this has not been tested within the present framework's specific stage structure.

Third, the framework as presented treats each cycle as a single pass through four stages, but real human-AI collaboration often involves multiple concurrent tasks and interleaved cycles at different timescales. How the framework should handle overlapping cycles, for example when an operator is simultaneously calibrating trust for two different subsystems, or how the task-allocation formalisms reviewed in Section 2.5 (Ali et al., 2021, 2022) might extend to concurrent cycles, is left for future work.

We present this framework as a starting point for evaluators seeking to measure trust calibration longitudinally rather than at a single decision point, and we welcome feedback from the workshop on the questions above, in particular on which measurement combinations and which trust mechanisms are most promising to pursue next.

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